

Hybrid Integrated Tunable Laser with SiN Ring Resonators for Wide Wavelength Tuning and Locking

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Abstract We demonstrate a hybrid integrated tunable external cavity laser utilizing SiN micro-ring resonators for tuning and locking of wavelengths, achieving a tuning range of 82 nm, an output power exceeding 16 dBm, and a SMSR over 56 dB.

Introduction

Driven by the emergence of AI, 5G/6G communications, and high-capacity cloud services, the explosive growth of global data traffic has placed unprecedented demands on the capacity of optical fiber infrastructures. Extending the operational bandwidth of WDM to the C+L bands is a critical strategy to enhance the spectral efficiency of the existing fiber infrastructure and support a greater number of optical channels [1, 2]. High-performance light sources for these systems must exhibit high optical power, narrow linewidth, and low relative intensity noise to support advanced high-order modulation formats in coherent optical communications [3, 4]. For these tunable light sources, silicon photonics (SiPh) technology has emerged as a transformative platform, taking advantage of CMOS-compatible manufacturing to provide compact, scalable, and low-cost solutions [5, 6]. Among various waveguide materials, silicon nitride (SiN) is particularly attractive for external cavity lasers (ECLs) due to its exceptionally low propagation loss and wide transparency window across the near-infrared spectrum. Due to the lack of native light emission in silicon, technologies such as heterogeneous integration via wafer bonding [1], hybrid integration through chip-to-chip butt-coupling [2], and micro-transfer printing [3] have been developed to combine III-V gain materials with low-loss SiN external cavities.

Achieving wide wavelength tunability typically relies on the Vernier effect, facilitated by two or more cascaded micro-ring resonators (MRRs) with slightly different radii to expand the effective free spectral range (FSR) [7, 8]. Furthermore, incorporating these resonators into a Sagnac loop reflector geometry substantially extends the effective cavity length. This enhancement is pivotal for suppressing the Lorentzian linewidth to the sub-100 Hz regime for ultra-low noise operation. Consequently, the device demonstrates a wide 118 nm tuning range, which seamlessly covers the S, C, and L bands [9]. High-power output is often achieved by integrating booster semiconductor optical amplifiers, with reported on-chip powers exceeding 20 dBm [6, 7].

In this work, we demonstrate a widely tunable external cavity laser (ECL) with SiN MRRs for tuning and locking of wavelengths. By employing hybrid integration by chip-to-chip edge-coupling, the proposed tunable laser achieves a wide wavelength tuning range, high output powers, and a narrow laser linewidth.

Design

The tunable ECL comprises SiN MRRs in a single chip for tuning and locking wavelengths, a commercial RSOA, and photodiodes (PDs) for optical power monitoring, as shown in Fig. 1. Through rigorous calculations of waveguide modes and coupling ratios between the MMR structure and the

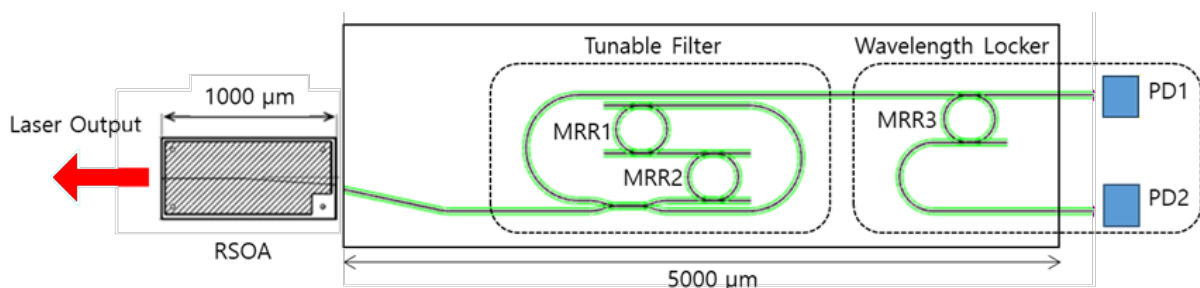


Fig. 1: Schematic of the hybrid integrated tunable laser with SiN MRRs for tuning and locking of wavelengths.

linear add-drop waveguides, the waveguide dimensions were optimized to a thickness of $0.4\ \mu\text{m}$ and a width of $0.8\ \mu\text{m}$. To minimize internal cavity losses, a totally reflective Sagnac loop mirror structure was implemented. This specific configuration ensures that the optical path only passes through the filter once within the cavity, effectively reducing the overall loss. The tunable filter consists of a dual-ring resonator structure with a free spectral range (FSR) of approximately 200 GHz, where the two rings are designed with a cavity length difference of $20\ \mu\text{m}$ to utilize the Vernier effect. To ensure a side mode suppression ratio (SMSR) sufficient for robust filtering, the optical coupling ratio between the bus waveguide and the ring resonators was precisely tuned to 15%.

Wavelength tuning is achieved by inducing a phase shift in the ring cavities through the thermo-optic effect, utilizing heaters deposited on top of the SiN MRRs to change the refractive index. For a heater-induced temperature change of approximately $50\ ^\circ\text{C}$, the tunable filter provides a tuning range corresponding to the Vernier FSR, $\Delta\lambda_V$, which is achieved through the synthesized resonance of the two rings with slightly different radii, R1 and R2, as illustrated in Fig. 1. In operation, the resonance wavelength is tuned in opposite directions by the two heaters. Heating MRR1 leads to a redshift in the peak wavelength, whereas heating MRR2 induces a corresponding blueshift. By applying differential heating and cooling, wide wavelength tuning is achievable within a phase shift range of $0.1\Delta\lambda_V$ to $0.4\Delta\lambda_V$. To suppress the spectral sensitivity of the directional coupler (DC) between the bus waveguide and the ring, a curved coupling structure was employed to maintain a consistent coupling coefficient in a wide wavelength range. This optimization resulted in a stable coupling ratio with a variation of only 1% across full C+L band.

Conventional wavelength lockers generally rely on external Etalon filters and multiple photodiodes (PDs), necessitating bulky components such as partial mirrors and thermo-electric coolers (TECs) that hinder miniaturization. To facilitate miniaturiza-

tion, this work demonstrates the monolithic integration of an MRR-based wavelength locker directly onto a SiN tunable filter platform. The wavelength locker utilizes a ring filter with a 100 GHz FSR aligned to the ITU grid. The through and drop ports of the locker ring, MRR3 are assigned to the reference and signal PDs, respectively, to monitor the spectral response for wavelength stabilization. The coupling ratio of 1/2 for the locker ring was set to ensure stable locking. While a high Q-factor is generally desirable, an excessively high value can make wavelength locking difficult if the signal intensity drops too sharply when off-resonance. Precise wavelength locking was successfully attained by fine-tuning each channel through heater-controlled resonance adjustment.

Fabrication

The SiN planar lightwave circuit (PLC) incorporating the MRRs was fabricated on a silicon substrate with a $5\text{-}\mu\text{m}$ -thick thermal oxide under cladding. To accommodate the $0.4\text{-}\mu\text{m}$ -thick stoichiometric SiN core deposited by low-pressure chemical vapor deposition (LPCVD) while mitigating the inherent tensile stress, a $0.3\text{-}\mu\text{m}$ -deep stress release pattern was pre-etched into the thermal oxide. This technique effectively suppresses crack formation, which typically occurs in SiN films exceeding $0.3\ \mu\text{m}$ in thickness. Following waveguide patterning, a $2.5\text{-}\mu\text{m}$ -thick SiO_2 upper cladding was deposited using plasma-enhanced chemical vapor deposition (PECVD). Finally, $0.5\text{-}\mu\text{m}$ -thick and $5\text{-}\mu\text{m}$ -wide aluminum heaters were integrated to allow efficient thermal adjustment of the device.

To optimize fiber-to-chip coupling efficiency, a spot-size converter (SSC) with a tip width of $0.3\ \mu\text{m}$ is integrated in the titled waveguide of the SiN PLC, facilitating efficient mode expansion. The SiN PLC was directly butt-coupled to the $1000\text{-}\mu\text{m}$ -long reflective semiconductor optical amplifier (RSOA) with a gain peak near $1580\ \text{nm}$, an antireflection-coated facet and a tilted coupling waveguide. The SSC of the SiN PLC is optimally designed to be tilted at 12 degrees for effective coupling into the RSOA to realize chip-to-chip bonding integration. The wavelength locker architecture was realized

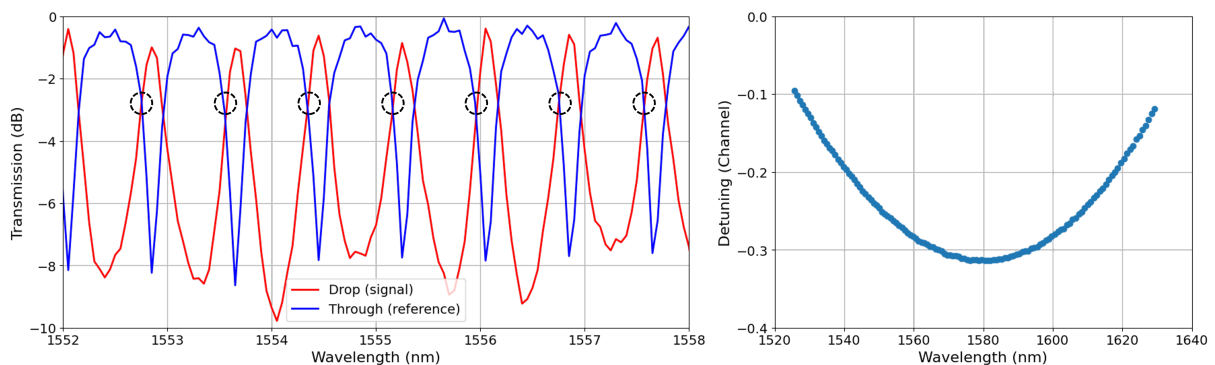


Fig. 2: (a) Transmission spectra of the wavelength locking ring filter and (b) normalized detuning.

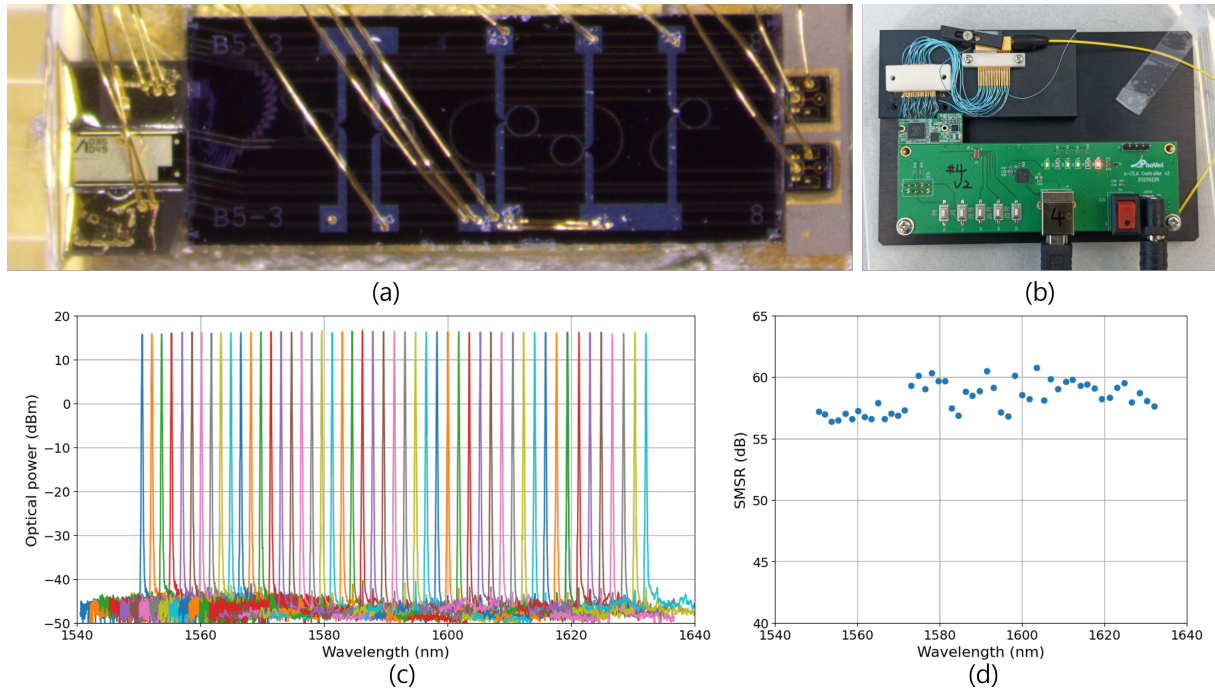


Fig. 3: Photograph of hybrid integrated ECL platform (a) and μ -iTLA connected to a controller, and measured output spectra (c) and laser linewidth (d) of the μ -iTLA.

by mounting the PD1 and PD2 for monitoring the signal and reference onto a glass-based carrier featuring pre-patterned electrodes. Then, this PD-integrated carrier was directly interfaced with the SiN waveguide platform, enabling a compact and robust locking module.

Results

The wavelength locker taps a portion of the light from the ECL cavity into the input of a ring filter, where the through and drop ports are monitored by reference and signal PDs, respectively. Fig. 2(a) shows the measured transmission spectra of the fabricated ring filter for the wavelength locker. The red and blue curves represent the drop (signal) and through (reference) ports, respectively, while the circles indicate the intersection points of the two outputs, marking the target wavelength locking positions. Fig. 2(b) presents the detuning between the ITU grid and the ring filter peaks, normalized to the channel spacing. The precise wavelength locking for each channel was established through heater-controlled thermal tuning.

Fig. 3(a) shows the hybrid integrated ECL utilizing SiN MRRs for tuning and locking of wavelengths, the RSOA, and the monitoring PDs. Detailed characterization of the fundamental ring resonators was conducted, showing a bus-to-ring coupling ratio of 18% and a propagation loss of 1.2 dB. Based on these elemental properties, the fully integrated tunable filter achieved a competitive insertion loss of 2 dB and an SMSR of approximately 1.8 dB, confirming its robust spectral performance. When integrated with the RSOA at a coupling dis-

tance of 5 μ m, the device demonstrated threshold currents of 20 to 25 mA, optical output powers of more than 8 dBm at 80 mA, and SMSRs exceeding 50 dB. The μ -iTLA (micro-Integrable Tunable Laser Assembly) was realized within a compact 20-pin ceramic feed-thru package with overall dimension of 14 mm $\times$ 5.6 mm $\times$ 4 mm. Optical output was coupled into a high-NA fiber through a glass ferrule. This μ -iTLA has been controlled to tune and stabilize the wavelengths by the evaluation board, as shown in Fig. 3(b). The μ -iTLA achieved a total wavelength tuning range of 82 nm, delivering an output power of more than 16 dBm while maintaining SMSRs over 56 dB, as shown in Figs. 3(c) and (d). The laser linewidth of less than 2 kHz was measured using the delayed self-heterodyne method.

Conclusions

We have successfully demonstrated a high-performance hybrid integrated μ -iTLA utilizing a SiN waveguide platform. Experimental results show high performance, including a wide tuning range of 82 nm, a high optical output power exceeding 16 dBm, and an SMSR of more than 56 dB. The monolithic integration of an MRR-based wavelength locker onto a SiN tunable filter platform enabled highly effective wavelength locking, achieving precise alignment to the 100 GHz FSR ITU grid while significantly advancing device miniaturization.

Acknowledgements

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